

Q1) We have demand curve for a product is given by

$$Q = 300 - 2P + 4I$$

I is average income measure in thousands of dollars.

Supply curve = $Q = 3P - 50$.

If $I = 25$, find market clearing price.

Sol:-

$$300 - 2P + 4I = 3P - 50$$

$$5P = 300 + 50 + 4(25)$$

$$P = \frac{\cancel{3750}}{\cancel{5}} \boxed{90}$$

Q2) A vegetable fiber is traded in competitive world market & world price is \$9 per pound.

Unlimited quantities are available

for import into the United States
at this price. The US domestic
supply and demand for various
price levels are shown as follows.

price	US supply	US demand
3	2	34
6	4	28
9	6	22
12	8	16
15	10	10
18	12	4

So⁽⁹⁾ $\left(\frac{y_2 - y_1}{x_2 - x_1} \right)$ $\frac{6-3}{4-2} = \left(\frac{3}{2} \right)$

$$y = \left(\frac{3}{2} \right) x + c$$

(12)

$$Q = \left(\frac{3}{2}\right)P^3 + C$$

} supply curve.

$$C=0$$

$$y = \frac{3}{2}x$$

$$\left(\frac{y_2 - y_1}{x_2 - x_1}\right)$$

$$Q = A + BP$$

$$\frac{22 - 28}{9 - 6}$$

$$= -6/3 = -2$$

$$y = mx + c$$

$$y = (-2)x + c$$

$$10 = -30 + c$$

$$c = 40$$

$$Q = -BP + A$$

$$y = -2x + 40$$

} demand

- 3) In a particular year, people consume 315 chocolates (or) 15.75 packets of chocolates. The avg retail price was about ~~₹25~~ ₹25 per packet. Suppose the price elasticity of demand is -0.4 and price $\dots$ is 0.5 .

elasticity of supply is -

Derive linear supply & demand curve.

Soln

$$\text{demand} = -0.4$$

$$\text{supply} = 0.5$$

$$Q = A - BP$$

$$-0.4 = \left(\frac{\Delta Q}{\Delta P} \right) \times \frac{5}{15.75}$$

$$\frac{\Delta Q}{\Delta P} = -1.26$$

$$15.75 = A - 1.26(5)$$

$$A = 15.75 + 1.26(5)$$
$$15.75 + 6.3$$

$$A = 22.05$$

$$Q = A + BP$$

$$0.5 = \left(\frac{\Delta Q}{\Delta P} \right) \frac{P}{Q}$$

$$0.5 = \left(\frac{\Delta Q}{\Delta P} \right) \times \frac{5}{15.75}$$

$$10.575 = \frac{\Delta Q}{\Delta P}$$

$$15.75 = A - 1.525(5)$$

$$A = 23.625$$

4)

<u>Year</u>	price of good 1	sales of good 1	price of good 2	sales of good 2
Year 1	10.35	75	4.11	820
Year 2	10.48	70	3.76	850

Using the table, find the price elasticity of demand for good 2. & derive linear demand curve.

soln

$$\frac{\Delta Q}{\Delta P}$$

$$m = \frac{850 - 820}{3.76 - 4.11}$$

$$\text{elasticity} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

5) Anand purchases either tables or bats.
Given by his utility function,

$$U(t, b) = 15t + 10b$$

Price of bat = 2

Price of tables = 0.5

Budget = ₹5. & He is planning to
buy 1 bat & 6 tables.

(Sol)

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

∴ Maximising
utility

$$\begin{array}{|l} x = \text{tables} \\ \hline y = \text{bat} \end{array}$$

$$\frac{15}{P_x} \neq \frac{10}{P_y}$$

$$\frac{15}{0.5} \neq \frac{10}{P_y}$$

$$\frac{10}{2} = 5$$

So, not maximising utility

Q. A consumer receives utility by consuming food (F) and clothing (C)

$$U(F, C) =$$

$$F = 2\text{£} \quad C = 10\text{£}$$

$$\text{Income} = 50\text{£}$$

Marginal rate of substitution of food for clothing when utility is maximised.

So 1)

$$\frac{P_x}{P_y} = \frac{2}{10} = \left(\frac{1}{5}\right) = 0.2$$

↑
slope of budget
line

= MRS when

max utility.

Pyndic & Rubin field

Dominic & Salvatore

Schawn's

